# AquaNavAI: A Reproducible Platform for Forecast-Aware Coastal ASV Route Planning Using NOAA-Derived Data

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***Abstract*—Coastal autonomous surface vehicle (ASV) route planning couples discrete search with bathymetric constraints, time-varying currents, vehicle assumptions, and data-provenance requirements. This paper presents AquaNavAI, an offline-capable, checksum-audited research platform that transforms pinned NOAA-derived inputs into a shared navigation grid, computes four route objectives, and exposes the same artifacts through a Python evidence pipeline and a browser demonstration. The released exploratory V1 case contains three missions at ten hourly forecast-valid times from one RTOFS initialization, yielding 30 matched cases and 120 routes. All 120 reconstructed route constraints passed with zero recorded violations; however, V1 uses a fixed 179.985 W proxy, so modelled energy is proportional to transit time, and no independent energy–time or planner-superiority claim is made. A prospective V2 protocol added a pre-acquisition convergence gate. Five synthetic checks passed, but only 27 of 30 excluded-pilot comparisons passed because one long mission was forecast-infeasible for three objectives at both tested arrival-label resolutions. The criterion was not satisfied; confirmatory acquisition was not initiated, and eligible cycles, cases, and routes remained zero. The contribution is a reproducible system and claim-control workflow, not a confirmatory planner-performance result. *research demonstration only; not for navigation*.**

***Index Terms*—autonomous surface vehicle, coastal route planning, environmental data, reproducibility, NOAA, A*, Dijkstra, modelled energy**

## I. Introduction

Autonomous surface vehicles (ASVs) support persistent sensing, mapping, and survey tasks, but their guidance stack must operate in an environment whose navigable space and forcing vary in space and time. Reviews of unmanned surface vehicles identify guidance, navigation, control, endurance, and environmental robustness as coupled system challenges rather than separable software features [1]. In coastal settings, a geometrically short polyline can cross shallow or invalid cells, while an apparently modest current can make an edge slow or infeasible for a reference vehicle.

Environmental route-planning demonstrations also face an evidence problem. Source products can be replaced, forecast-valid times can be mislabeled as independent initializations, and figures can outlive the data-processing assumptions that created them. Reproducible computational practice therefore requires recorded inputs, transformations, versions, and executable reconstruction steps [2]; FAIR principles extend that concern to data, algorithms, tools, and workflows [3].

AquaNavAI addresses these issues as a research platform rather than an operational navigator. Its contributions are:

1) a pinned, checksum-audited NOAA-derived South Florida scenario with explicit source roles, units, transformations, and exclusions;
2) four route objectives on a common graph, forecast, feasibility mask, and reference-vehicle model;
3) an offline browser architecture that performs route computation in a persistent Web Worker and visualizes routes and ASV motion; and
4) an evidence package that separates exploratory V1 observations from a prospective V2 protocol whose pre-acquisition criterion prevented confirmatory acquisition when declared invariants could not be established.

The paper intentionally makes no claim of a trained machine-learning model, planner superiority, globally optimal continuous-water routing, measured vessel energy, or operational safety.

## II. Related Work

Dijkstra's method provides the canonical nonnegative-edge shortest-path baseline [4]. A* augments accumulated cost with a heuristic and can retain minimum-cost guarantees on its declared graph under the classical conditions [5]. Such guarantees apply to the chosen graph, state, and cost model; they do not establish global optimality in continuous coastal waters.

Environmental USV routing has coupled graph construction, current forcing, and energy models in several ways. Niu *et al.* combine Voronoi and visibility graphs with Dijkstra search, an energy function, sea-current data, and configurable coastline clearance [6]. Zhang *et al.* formulate dynamic energy-efficient planning under time-varying current and wind using an A*-based approach [7]. These studies motivate environmentally aware costs, whereas the present V1 evidence uses constant through-water speed and a fixed-power proxy; its energy values are neither measured nor independent of transit time.

Current and obstacle constraints also appear in improved graph and hybrid search. Yang *et al.* add current-dependent potential-field terms to an improved A* planner and subsequently smooth paths [8]; Sang *et al.* combine improved A* with a multi-subtarget artificial potential field for USV formations [9]. Enevoldsen *et al.* incorporate COLREGs information into RRT* for marine-craft collision avoidance [10]. In contrast, AquaNavAI preserves exact discrete grid nodes, applies no route smoothing, and excludes traffic, dynamic obstacles, and COLREGs compliance; it therefore makes no collision-avoidance or operational-navigation claim.

Coastal work must also represent water depth and survey boundaries. Wilson and Williams demonstrate adaptive, depth-constrained bathymetric mapping with online Gaussian-process updates and field tests on an autonomous surface vessel [11]. The present platform has a narrower role: pinned CUDEM topobathymetry defines a released feasibility mask, but the 500 m grid is not an active chart and cannot certify navigability.

The system contribution is reproducibility and claim control across acquisition, processing, planning, interface, and evidence. Reproducible-computation guidance emphasizes recorded inputs, transformations, versions, and executable steps [2], while FAIR principles apply to data, algorithms, tools, and workflows [3]. AquaNavAI accordingly pins and checksum-verifies environmental inputs, deploys same-origin artifacts offline, audits route reconstruction, and packages provenance, UI demonstration, and research evidence. It does not propose a universally optimal planner.

## III. Platform Architecture

Figure 1 shows the deployment and evidence boundary. Explicit Python acquisition adapters retrieve source products; processing harmonizes coordinates, masks, currents, and metadata into bounded scenario artifacts. The released web application then operates only on same-origin static files. There is no V1 backend API and no runtime NOAA dependency.

The frontend uses React and MapLibre GL JS [12]. A persistent Web Worker decodes the navigation grid once, correlates route requests, rejects superseded work, and returns all four objectives. Separating the worker keeps graph search off the interface thread. The map renders raw planner nodes, currents, mission endpoints, and a 42-pixel HTML vessel marker. A single animation loop interpolates cumulative geodesic distance along the selected polyline; the 22-second demonstration clock is visibly distinct from modelled transit time.

The static deployment includes locally bundled map styling, scenario layers, metrics, attribution, and provenance. Thus the interactive research demonstration can run with external networking blocked. Numerical values in this paper are read from machine-readable artifacts, never from screenshots.

## IV. Environmental Data and Provenance

The released domain spans longitude $[-80.24, -80.005]$ and latitude $[25.55, 26.70]$. WGS84 (EPSG:4326) is used for interchange; EPSG:32617 is used for the 500 m analysis grid. Table I summarizes source roles.

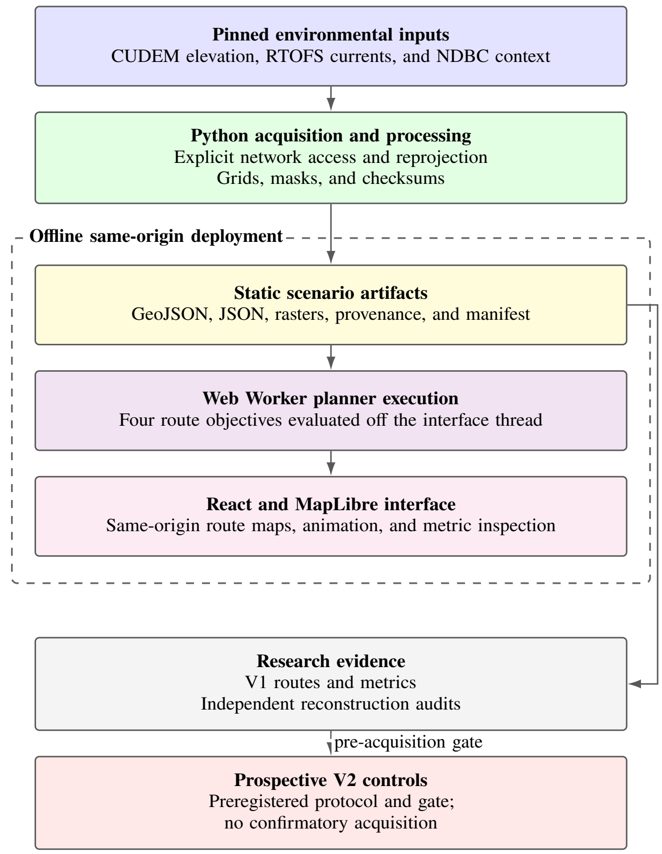


Fig. 1. Platform architecture and evidence flow. Network access occurs only during explicit acquisition and processing. The dashed boundary encloses the offline same-origin deployment. Static scenario artifacts support both browser-side route computation and independently reconstructed research evidence. The prospective V2 gate did not initiate confirmatory acquisition.

CUDEM Florida 1/9 arc-second tiles supply elevation, land, shallow-water context, and finite-coverage constraints [13]. Five source tiles were retrieved on 2026-07-12 and individually checksummed. Elevation was bilinearly reprojected, depth was defined as $d = \max(-z, 0)$, feasible cells required $d \geq 5$ m, and the configured 250 m land/NoData buffer rounded to one 500 m cell. CUDEM is not a navigation chart and its finer source spacing does not remove the limitations of the analysis grid.

The current field is the RTOFS western Atlantic standard-field product, which is based on the HYCOM ocean-model family [14], initialized 2026-07-12 00Z, with eastward and northward surface velocity in m/s [15]. Components were independently reprojected to the shared grid. The source contains hourly valid fields through lead hour 24; V1 routes use lead hours 1–10. These are ten valid times from one initialization, not ten independent cycles.

NDBC stations 41122 and LKWF1 provide point meteorological and wave context; NDBC describes these real-time files as automatically quality-controlled recent records [16]. They do not validate the spatial RTOFS current field. NOAA ENC

TABLE I
NOAA-derived inputs and release status. Native resolution is retained in provenance; interpolation does not create new information.

| Source | Product/status | Temporal or edition scope | Native support | Released role and processing |
|---|---|---|---|---|
| CUDEM | Florida 1/9 arc-second topobathymetry; pinned | Tile editions 2016v1 and 2018v1 | Approx. 3 m; elevation (m) | Bilinear reprojection to 500 m EPSG:32617; depth/feasibility source |
| RTOFS | Western Atlantic standard fields; pinned | 2026-07-12 00Z; hourly valid 01Z–24Z | 0.08 degree, 575×435; surface $u, v$ (m/s) | Component-wise bilinear reprojection; modelled current forcing |
| NDBC | Stations 41122 and LKWF1; pinned | Latest records retrieved 2026-07-12 | Point observations; header-defined units | Context only; not spatial current validation |
| NOAA ENC | Electronic Navigational Charts; audited | No chart cells integrated | Vector chart features | Future hazards/restrictions source; no V1 mask effect |
| WAVEWATCH III | Wave model; deferred | Not acquired | Not acquired | No wave value enters V1 routing |

was audited as a future chart-constraint source [17], but no chart feature changes the V1 mask. WAVEWATCH III is also deferred [18]; no wave value enters any reported cost.

The scenario manifest checksums nine released artifacts. Provenance records source URLs, retrieval times, products, variables, units, source and processed coordinate systems, temporal support, transformations, and limitations. The browser validates the contract, while the research package independently reconstructs route geometry against the grid rather than trusting displayed metrics.

## V. Route-Planning Methodology

The shared navigation grid has 49 by 256 cells with eight-neighbor connectivity. Let an edge of length $\ell$ have unit along-track vector $\hat{e}$ and modelled current $\mathbf{c}$. The current is decomposed into along-track and cross-track components $c_\parallel$ and $c_\perp$. For through-water speed $v = 1.5$ m/s, an edge is infeasible if $|c_\perp| \geq v$. Otherwise its forward ground speed is

$$v_g = \sqrt{v^2 - c_\perp^2} + c_\parallel. \tag{1}$$

The edge is also infeasible when $v_g \leq 0$; feasible travel time is $t_e = \ell / v_g$. Waiting and forecast extrapolation are disabled.

The reference model uses hotel power $P_h = 30$ W and

$$P_p = 44.44 v^3 \text{ W}, \qquad E_e = (P_h + P_p) t_e / 3600 \text{ Wh}. \tag{2}$$

At the fixed speed, total power is approximately 179.985 W. Consequently, V1 modelled energy is a rescaling of travel time, not an independently calibrated propulsion endpoint.

Shortest distance uses Dijkstra on static feasible edges; currents are applied only when its released geometry is evaluated. The other objectives use deterministic time-expanded A* states $(i, j, k)$, where $k$ is the one-hour forecast bin. Fastest minimizes accumulated $t_e$; lowest modelled energy minimizes accumulated $E_e$; balanced combines normalized time, energy, current exposure, and shallow-water context with the weights in Table II. The production implementation uses a zero heuristic for these three objectives, retaining admissibility on the declared discrete state graph. No claim is made about a continuous-water global optimum.

Because the V1 fixed-power energy proxy is mathematically proportional to transit time, the time and energy terms in the balanced objective are not independent criteria. Consequently, V1 balanced routing partially double-weights the same underlying transit-time signal and must not be interpreted as a genuine independent four-factor trade-off.

## VI. Exploratory V1 Evaluation

V1 is a fixed-case exploratory evaluation. It contains one RTOFS initialization, ten hourly forecast-valid times, three missions, and four planners. The 30 matched mission–time cases therefore yield 120 route records, as shown in Table III. The ten times share one environmental initialization and are not independent forecast cycles.

For every route, the evidence extractor snapped each GeoJSON coordinate to the nearest released grid node and checked snap error, feasible-water membership, minimum depth, neighbor adjacency, forecast availability, current feasibility, path-length agreement, and minimum-depth agreement. All 120 routes passed with zero recorded violations. This establishes consistency with the released discrete model; it does not establish real-world navigational validity.

Table IV reports descriptive values recomputed from the tidy artifact. Mean transit time was 16262.9 s for shortest and 16200.3 s for fastest and lowest-modelled-energy routes, a fixed-case difference of about 62.6 s. The balanced mean was 16200.3 s after rounding. These observations are not a planner ranking: missions and valid times are deliberately narrow, and fastest and energy coincide under fixed power.

Figure 3 shows paired percentage differences without treating the 30 cases as independent environmental replicates. Figure 4 shows within-initialization variation in the energy difference and states the proportionality limitation directly. No Pareto frontier is reported because the energy and time axes contain the same information up to constant scaling.

Computation-time fields contain source placeholders and an explicit availability flag of false. Runtime evidence was not measured (0 observations); no runtime comparison, plot, test, or performance claim is included. This unavailable evidence category is distinct from software verification status.

## VII. Prospective V2 Design and Gate Outcome

Experiment V2 was designed prospectively to replace V1's single-initialization scope with ten eligible consecutive 00Z RTOFS initializations, ten missions, and four planners. Its frozen target was 100 matched cases and 400 routes, with

TABLE II

Implemented route objectives. All planners share endpoints, forecast field, vehicle, grid, and feasibility mask.

| Objective | Search | Cost minimized | Claim boundary |
|---|---|---|---|
| Shortest distance | Dijkstra | Geometric eight-neighbor edge distance | Static feasible grid |
| Fastest arrival | Time-expanded A* | Current-aware transit time | No waiting or forecast extrapolation |
| Lowest modelled energy | Time-expanded A* | Fixed reference-power energy proxy | Not measured vessel energy |
| Balanced mission | Time-expanded A* | Normalized time 0.35, energy 0.35, current exposure 0.15, shallow-water context 0.15 | Time and energy are proportional in V1; partially double-weights transit time; not an operational safety score |

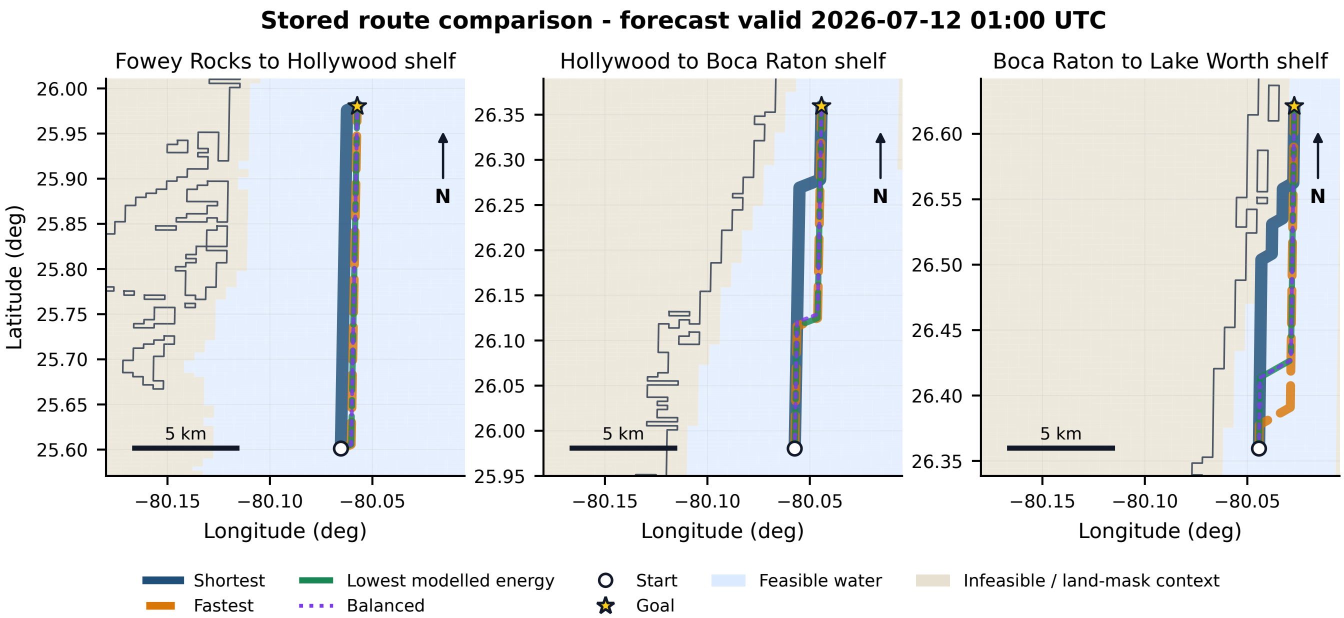


Fig. 2. All four stored planner outputs for each of the three released missions at 2026-07-12 01:00 UTC. The representative time is selected by the declared non-outcome-based rule “first released forecast-valid time.” Coordinates are the exact stored EPSG:4326 route vertices; no smoothing, offset, or jitter is applied. Widths and dash patterns expose coincident paths: overlapping lines indicate identical or nearly identical discrete grid routes and do not mean that hidden planners were omitted. Background cells show the released feasible-water mask and infeasible/land-mask context; the coastline is the released NOAA-derived layer.

TABLE III

Exploratory V1 evidence inventory. Runtime is an intentionally unavailable evidence category, not a software gate outcome.

| Quantity | Evidence |
|---|---|
| RTOFS initializations | 1 |
| Forecast-valid times | 10 |
| Missions | 3 |
| Matched mission–time cases | 30 |
| Planners / routes | 4 / 120 |
| Valid routes / violations | 120 / 0 |
| Runtime evidence | Not measured (0 observations); no runtime comparison claimed |

separate runtime, memory, and vehicle-parameter sensitivity protocols. Those targets remain plans, not observations.

Before acquisition, the protocol required the production 300 s arrival-label resolution to agree with a 60 s reference on excluded V1 pilot inputs and deterministic synthetic fixtures. The audit prohibited confirmatory inputs and required both feasibility/constraint status and objective agreement within the declared tolerances. Table V reports the immutable outcome.

All five synthetic checks passed. Of 30 excluded-pilot comparisons (10 missions by three environmental objectives), 27 passed. Mission `m10-midshelf-south-long` returned no forecast-feasible route for fastest, energy, and balanced objectives at both 300 s and 60 s resolutions. Feasibility therefore matched, but no objective value existed; the declared constraint-status and cross-planner invariant set could not be established. The audit correctly did not convert matching missing objectives into convergence evidence.

The gate criterion was not satisfied, so confirmatory acquisition was not initiated. Eligible cycles/cases/routes remained 0/0/0, and runtime, memory, and sensitivity observations remained 0/0/0. Proposed Amendment 0004 records the gate outcome but is not adopted. This is validation-gate evidence only; there are no confirmatory statistical findings and no basis for a planner ranking.

The exact local paths and SHA-256 values are preserved in the manuscript claim audit. The gate manifest begins `d5d0d412fc03`; its full identifier binds this wording to the immutable comparison, synthetic, report, and attempt records

TABLE IV
EXPLORATORY FIXED-CASE V1 DESCRIPTIVE STATISTICS. EVERY NUMERIC CELL IS MEAN ± SAMPLE SD ACROSS $n$ = 30 ROUTES PER PLANNER. ENERGY IS MATHEMATICALLY PROPORTIONAL TO TIME UNDER THE FIXED 179.985 W PROXY.

| Planner | Modelled energy (Wh) | Transit time (s) | Path length (m) | Minimum depth (m) |
|---|---|---|---|---|
| Shortest | 813.08 ± 112.77 | 16262.9 ± 2255.6 | 38080.9 ± 6084.7 | 61.64 ± 42.46 |
| Fastest | 809.94 ± 110.90 | 16200.3 ± 2218.2 | 38080.9 ± 6084.7 | 64.08 ± 43.02 |
| Lowest modelled energy | 809.94 ± 110.90 | 16200.3 ± 2218.2 | 38080.9 ± 6084.7 | 64.08 ± 43.02 |
| Balanced | 809.95 ± 110.91 | 16200.3 ± 2218.3 | 38080.9 ± 6084.7 | 64.08 ± 43.02 |

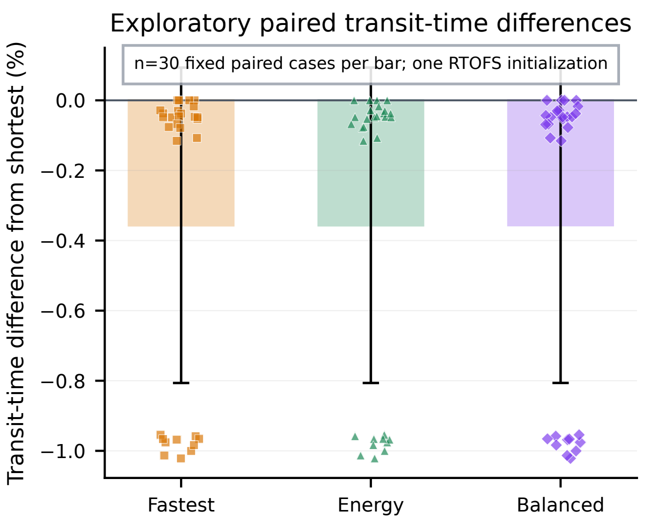


Fig. 3. Exploratory paired transit-time differences from the shortest route. Each colored bar is the mean percentage difference for one planner across 30 fixed mission–valid-time cases; its black whisker is ± one sample SD, not a confidence interval. Each marker is one paired case (30 observations per bar). All cases come from three missions and ten valid times within one RTOFS initialization.

TABLE V
PRE-ACQUISITION CONVERGENCE-GATE OUTCOME.

| Check | Outcome |
|---|---|
| Synthetic fixtures | 5/5 passed |
| Excluded-pilot comparisons | 27/30 passed |
| Unresolved comparisons | `m10-midshelf-south-long`; fastest, lowest modelled energy, balanced; infeasible at 300 s and 60 s label resolutions |
| Confirmatory evidence | 0 cycles / 0 cases / 0 routes |
| Gate decision | Gate criterion not satisfied; confirmatory acquisition was not initiated |

without treating the gate as a scientific outcome.

## VIII. SOFTWARE VERIFICATION

The recorded application release is version 0.2.0. Exact tool versions, source identifiers, and artifact hashes are retained in the reproducibility audit rather than repeated here.

Table VI summarizes the recorded release gate. Python checks cover formatting/lint, strict typing, unit/integration tests, scenario validation, and dependency audit. Frontend checks cover TypeScript, ESLint, Vitest, the Vite production build, static-asset limits, and offline Playwright behavior. The browser suite blocks external requests, checks desktop/mobile rendering

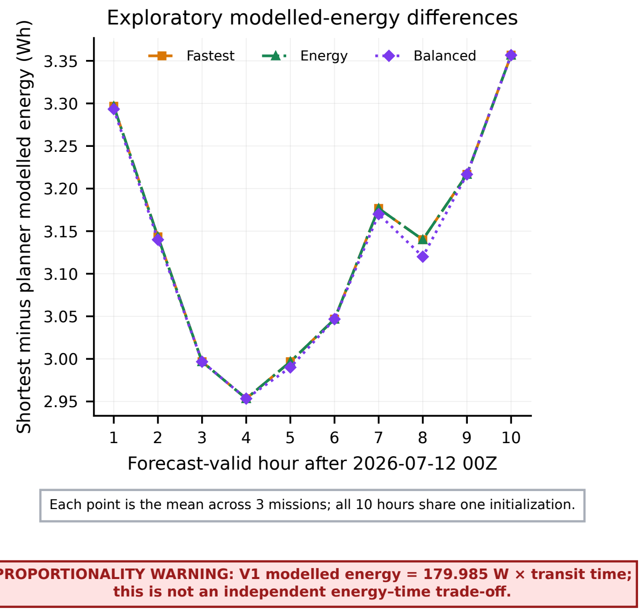


Fig. 4. Exploratory V1 modelled-energy differences by forecast-valid hour. Each point is the arithmetic mean of three mission observations ($n$ = 3) for that hour, computed as shortest minus the named planner. The ten hourly points come from one RTOFS initialization, not ten independent initialization cycles. Fixed-power proportionality makes this a transit-time transformation, not an independent energy result.

TABLE VI
RECORDED LOCAL RELEASE VERIFICATION.

| Gate | Recorded outcome |
|---|---|
| TypeScript, ESLint, Vitest, Vite, assets | Pass; 6 unit/component tests |
| Ruff and strict mypy | Pass |
| Python pytest | 21 passed |
| Scenario checksums | 9 artifacts passed |
| Playwright Chromium | 2 offline desktop/mobile tests passed |
| Dependency audits | No known vulnerabilities |

and controls, exercises custom-route and invalid-land behavior, and detects console, page, and network errors.

These checks establish bounded static deployment, not route-computation benchmarks. Separately, the research package verifies artifact hashes and reconstructs route constraints from machine-readable geometry. Software correctness, artifact integrity, and scientific generalization remain distinct evidence categories.

TABLE VII

CLAIM BOUNDARIES FOR THE PRESENT SYSTEM/REPRODUCIBILITY PAPER.

| Topic | Supported by existing evidence | Not supported |
|---|---|---|
| Planner behavior | Four implemented objectives; fixed-case route outputs | Superiority or global optimality |
| Energy | Modelled fixed-power proxy | Measured propulsion or independent energy–time Pareto |
| Forecast sample | Ten valid times from one initialization | Ten independent cycles |
| Software | Recorded tests, checksums, offline demonstration | Operational navigation or collision avoidance |
| V2 | Prospective protocol and failed pre-acquisition gate | Confirmatory outcomes |

## IX. DISCUSSION

AquaNavAI demonstrates why a research platform benefits from treating claim control as part of system design. The same-origin scenario boundary makes the public demonstration repeatable and prevents network availability from silently changing a route. Checksummed source metadata and processed artifacts make data lineage inspectable. Independent reconstruction of route constraints reduces reliance on the UI or on metrics written by the planner itself.

The V1 evidence illustrates the difference between a valid software output and a general scientific conclusion. Every reconstructed route satisfied the released grid constraints, but the environmental sample consists of only one RTOFS initialization. Likewise, a modelled-energy column can be internally correct while remaining scientifically non-independent from time because the proxy power is constant. Preserving that limitation is more informative than presenting a degenerate Pareto plot.

The V2 gate provides a second distinction. Its failure is not a failed planner comparison; confirmatory comparison never began. The long pilot mission exceeded the forecast-feasible search conditions for three objectives at both label resolutions. That observation may reflect the mission, horizon, constraint model, or their interaction, but the frozen gate did not permit post-outcome repair. The scientifically defensible action was to retain the evidence and stop before acquisition. A future protocol can revise feasibility handling prospectively, then run a new gate without reclassifying the current failure.

## X. LIMITATIONS AND FUTURE WORK

Table VII separates demonstrated properties from unsupported claims. The largest scientific limitation is the absence of confirmatory V2 evidence. V1 uses one initialization, three fixed missions, and a 500 m grid; its results cannot characterize other cycles, regions, or vehicles. The energy proxy omits calibrated hull, motor, sea-state, control, and hotel-load variation. Computation time was intentionally unavailable in the released evidence (0 measured observations), so no runtime comparison is claimed and controlled timing and memory benchmarks remain future work.

The feasibility mask is derived from CUDEM, not an active navigational chart. NOAA ENC hazards and restrictions are not integrated. Waves, tides, traffic, regulations, uncertainty propagation, dynamic obstacles, and collision avoidance are excluded. Interpolation does not increase RTOFS or CUDEM information, and NDBC stations are insufficient for spatial current validation. Accordingly, no output is operationally safe or navigation-authoritative.

Future work should adopt a reviewed prospective amendment or a new protocol that defines treatment of forecast-infeasible missions before seeing confirmatory outcomes. After the gate passes, independent cycles can support cycle-level inference, controlled runtime/memory trials, grid and vehicle sensitivity, and cycle-aligned wave forcing. A learned energy or risk cost should be deferred until simulator- or platform-calibrated labels exist with grouped train/validation/test splits. Publication of detailed methods or engine code should also follow the project's IP/disclosure review.

## XI. CODE AND DATA AVAILABILITY

The public project repository is available at https://github.com/BipinChowdary/AquaNavAI. Application version 0.2.0 is identified by commit `5b58a7985c24a1b51b66998562a7b19f7d3688a9`; the V1 evidence-producing source state is `eaf617d27d0460a12b057fb18489ed73b3e59591`. The repository contains the machine-readable scenario, 120 route geometries, metrics, provenance, checksums, figure and table scripts, and reproduction instructions. The artifact is versioned by Git commit but has not been archived under a DOI.

## XII. CONCLUSION

AquaNavAI integrates NOAA-derived coastal data, a common constrained grid, four route objectives, provenance, verification, and an offline interactive demonstration. The exploratory V1 package provides 120 internally validated route records while clearly exposing its single-initialization and fixed-power limits. The prospective V2 workflow contributes a pre-acquisition gate whose criterion was not satisfied when convergence and cross-planner invariants could not be established; confirmatory acquisition was not initiated. The resulting contribution is a reproducible platform and evidence-boundary workflow, not evidence of route-performance improvement. *research demonstration only; not for navigation.*


## ACKNOWLEDGMENT AND AI-ASSISTANCE DISCLOSURE

This study, including the research concept, system architecture, experimental planning, implementation decisions, analysis, and scientific conclusions, was conceived and executed by Bipin Chandra Chowdary Vadlamudi. OpenAI Codex was used as a software-development and manuscript-revision aid. It assisted with iterative code refinement and automated verification; literature organization; LaTeX and editorial consistency checks; and proposed wording across the Abstract through Conclusion, the Code and Data Availability section, and this disclosure. The author directed, checked, and revised those contributions against the cited primary sources and repository evidence. Codex was not used to generate experimental observations, select outcomes, or determine the scientific conclusions.

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